NOVARION — a 3D coronary case-review system

Multimodal AI Hackathon 2026 · Track A: Cardiovascular Risk Visualization & Prediction · model v1.0.2 · **Decision support / educational purposes only. Not a substitute for formal diagnostic imaging.**

1. Problem, intended use and system

Risk percentages alone do not show *where* a model sees disease or *why*. NOVARION predicts the overall CAD label and the ≥50% stenosis labels of the LAD, LCX and RCA from routine clinical features, maps each vessel's calibrated probability onto an interactive 3D heart inside a translucent torso, and explains every prediction with exactly additive model contributions next to the patient's measurements. It is built as a review workflow (case → data-quality gate → prediction → 3D vessel inspection → explanation → human review note → hash-verified Review Snapshot → case history), not as a diagnostic tool: it never recommends tests or treatment, never localizes lesions within a vessel, and shows a persistent disclaimer on every screen and printout.

Architecture. UCI #411 (checksum-pinned) → registry-driven canonicalization + leakage guard → nested cross-validation (4 model families, 92 pre-registered configurations) → Platt calibration → two threshold layers → versioned, hash-verified model bundle (v1.0.2) → stateless FastAPI service (Pydantic contract → OpenAPI → generated TypeScript types) → React 19 + React Three Fiber client (Zustand state, IndexedDB case store). Registries (features, targets, anatomy map) make new features, models or structures a configuration change.

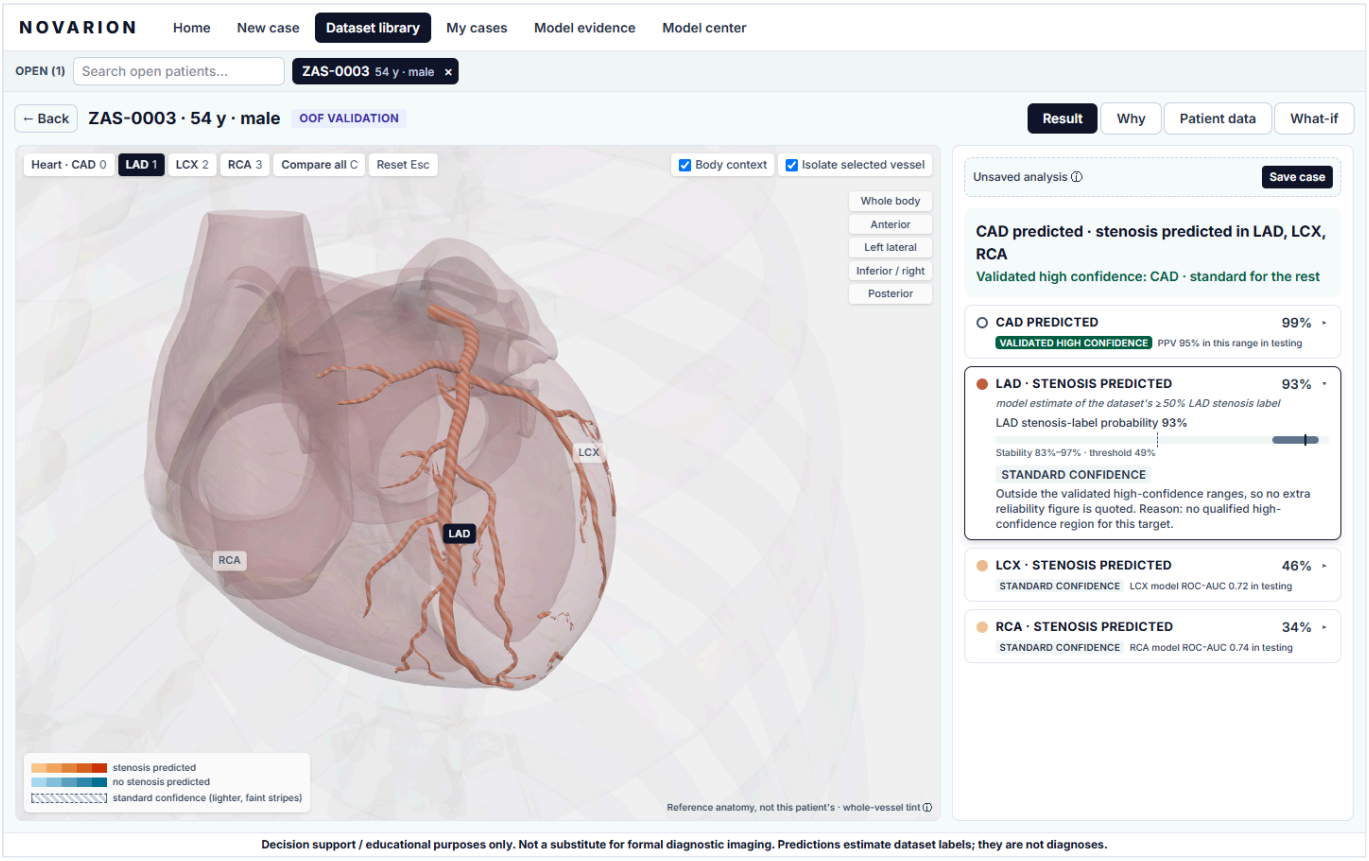


Figure 1 — A case's Result tab on a held-out dataset patient (out-of-fold prediction): LAD selected; the camera flies to the vessel and the others fade. Colour family = predicted class, depth = distance from the decision threshold, lighter with faint stripes = standard confidence. Beside it, a two-line summary and one line per target (binary call, probability, confidence) that expands to its stability interval and reasons.

2. Data and preprocessing

Extension of Z-Alizadeh Sani (UCI #411, CC BY 4.0), verified on download: 303 patients × 55 predictors, no missing values, no duplicate rows. Prevalence: CAD 71.3%, LAD 58.4%, LCX 39.3%, RCA 37.6%. The overall label is `Cath`; it agrees with "any vessel stenotic" in 302/303 records (one discordant record, kept as published, D-001). A feature registry fixes each feature's encoding, unit, valid range, UNKNOWN policy and lineage: binaries 0/1, ordinal VHD and function class, one-hot bundle-branch block and RWMA region (codes have no natural distance). BMI and obesity are computed from weight and height for new cases; a constant feature is auto-filled. **Leakage:** Cath, CAD, LAD, LCX and RCA (and any feature derived from them) are excluded from every model's inputs, enforced by a lineage-aware guard and tests. Every data-dependent step (zero-variance removal, log1p for skewed labs and scaling for linear models, imputation of explicit UNKNOWN values) is fit inside each training partition only.

3. Modelling and validation

Nested cross-validation per target: outer stratified 5-fold × 5 repeats (25 test folds); inner stratified 5-fold selects among 92 pre-registered configurations — logistic regression (L2/elastic-net), EBM without interactions, XGBoost and CatBoost (shallow), × class weighting × two feature sets. Selection uses inner-CV ROC-AUC with a 1-SE rule toward the simplest configuration; outer folds only estimate performance. **Calibration:** Platt scaling on cross-fitted logits (guard a > 0; never triggered). **Layer A** (official binary prediction): Youden threshold on calibrated training-side OOF probabilities. **Layer B** (model confidence): HIGH only inside a region

whose Wilson-95% lower bound meets the pre-registered PPV ≥ 0.90 / NPV ≥ 0.95 (n ≥ 15); otherwise LOW with a stated reason; UNKNOWN inputs and atypical profiles (Gower distance to the training cohort) force LOW. The app labels the tiers "validated high confidence" and "standard confidence". The deployed model per target repeats the selection on all 303 patients and refits. Logistic regression was chosen in most outer folds (Table 3), and in the deployed models for CAD, LAD and RCA (LCX: shallow XGBoost). **Explanations** are exact: linear SHAP (LR), TreeSHAP (XGBoost/CatBoost) or native EBM terms in the raw margin, scaled by the Platt slope into calibrated log-odds and summed per registry feature; base + Σ contributions = the displayed calibrated logit is tested for every served explanation. Dataset patients are always shown with their **out-of-fold** model (the fold that never saw them) and that fold's reference cohort. **Stability:** for each dataset patient, the fold's selected configuration is refit with Platt on 100 bootstrap resamples of that fold's training rows only (patient-grouped internal cross-fitting, threshold fixed); if the 10th–90th percentile interval straddles the decision threshold the patient is shown LOW with reason "unstable" (CAD 63, LAD 95, LCX 201, RCA 158 of 303; D-017). After every display gate the CAD HIGH badge covers 44.6% of patients at PPV 98.5% (Table 2b). Evidence-band ledgers use each fold's own inner cross-fitted scores, so a held-out patient's label never shapes its own evidence (patch release v1.0.2, D-019, found by an external audit).

4. Results

Table 1 — official metrics at the Layer-A threshold. Scope: outer-fold mean ± SD (25 folds). Italic rows: always-positive baseline (and the prior-probability Brier).

Target	Prev.	ROC-AUC	PR-AUC	Accuracy	Bal. acc.	Precision	Recall	F1	Brier	Cal. slope
CAD	71.3%	0.930 ± 0.035	0.971 ± 0.015	0.856 ± 0.050	0.860 ± 0.051	0.943 ± 0.034	0.851 ± 0.063	0.893 ± 0.041	0.095 ± 0.027	1.10 ± 0.63
		always positive	—	0.713 ± 0.007	0.500	0.713 ± 0.007	1.000	0.832 ± 0.005	0.205 ± 0.003	—
LAD	58.4%	0.838 ± 0.046	0.873 ± 0.042	0.771 ± 0.044	0.758 ± 0.044	0.789 ± 0.045	0.836 ± 0.081	0.809 ± 0.042	0.161 ± 0.023	1.00 ± 0.39
		always positive	—	0.584 ± 0.006	0.500	0.584 ± 0.006	1.000	0.737 ± 0.005	0.243 ± 0.001	—
LCX	39.3%	0.721 ± 0.067	0.601 ± 0.080	0.652 ± 0.066	0.666 ± 0.067	0.547 ± 0.071	0.731 ± 0.153	0.619 ± 0.083	0.209 ± 0.021	1.12 ± 0.53
		always positive	—	0.393 ± 0.005	0.500	0.393 ± 0.005	1.000	0.564 ± 0.006	0.239 ± 0.001	—
RCA	37.6%	0.736 ± 0.037	0.621 ± 0.057	0.655 ± 0.042	0.666 ± 0.042	0.536 ± 0.053	0.712 ± 0.127	0.605 ± 0.055	0.200 ± 0.012	1.07 ± 0.49
		always positive	—	0.376 ± 0.005	0.500	0.376 ± 0.005	1.000	0.547 ± 0.006	0.235 ± 0.001	—

Table 2 — confidence tiers. Scope: pooled OOF (prespecified repeat), patient-level bootstrap 95% CI.

Target	AUC [CI]	HIGH-pos.	PPV [CI]	HIGH-neg.	NPV [CI]	LOW	FP in HIGH-pos.
CAD	0.931 [0.898–0.957]	61.7%	94.7% [91.1%–97.7%]	0.0%	—	38.3%	10
LAD	0.813 [0.761–0.861]	0.0%	—	0.0%	—	100.0%	0
LCX	0.732 [0.674–0.785]	0.0%	—	0.0%	—	100.0%	0
RCA	0.732 [0.673–0.791]	0.0%	—	0.0%	—	100.0%	0

CAD is ranked well (ROC-AUC 0.930 ± 0.035) and 61.7% of patients fall in a qualified HIGH-positive region with measured PPV 94.7%. Vessel-level separation from routine features is weaker (LAD 0.838 ± 0.046, LCX 0.721 ± 0.067, RCA 0.736 ± 0.037): **no vessel model qualified a high-confidence region**, so every vessel prediction is shown with standard confidence (the LOW tier) while still giving a binary call, next to the model's tested ROC-AUC. The targets were fixed before results and were not lowered (D-008). Accuracy should always be read against the always-positive baseline (0.713 ± 0.007 for CAD).

Table 3 — family benchmarks (same selection rule within one family; ROC-AUC, outer-fold mean ± SD) and families selected across the 25 outer folds.

Target	Selected	LR	EBM	XGBoost	CatBoost	Selected family counts
CAD	0.930 ± 0.035	0.930 ± 0.035	0.933 ± 0.036	0.920 ± 0.038	0.930 ± 0.035	lr 24, ebm 1
LAD	0.838 ± 0.046	0.840 ± 0.045	0.839 ± 0.040	0.834 ± 0.042	0.827 ± 0.048	lr 22, xgboost 3
LCX	0.721 ± 0.067	0.717 ± 0.057	0.700 ± 0.065	0.736 ± 0.069	0.711 ± 0.065	lr 10, xgboost 13, ebm 2
RCA	0.736 ± 0.037	0.743 ± 0.045	0.717 ± 0.043	0.713 ± 0.049	0.717 ± 0.044	lr 23, xgboost 1, ebm 1

Against the guideline pretest score. Reviews of cardiovascular AI ask for comparison with established clinical scores, calibration beyond AUC and subgroup reporting (Parizad et al., Vasc Health Risk Manag 2026). For CAD we compared NOVARION with the updated Diamond–Forrester pretest probability (age × sex × chest-pain type; ESC 2013) on the 273 patients with a chest-pain type, using the same out-of-fold predictions.

Table 6 — CAD versus the guideline pretest score. Scope: pooled OOF (prespecified repeat), paired patient bootstrap 95% CI. Calibration-in-the-large = observed − mean predicted.

Model	AUC	Calibration-in-the-large	ECE	ΔAUC of NOVARION [95% CI]
NOVARION	0.935	+0.013	0.059	—
Updated Diamond–Forrester (published)	0.852	+0.158	0.168	+0.083 [0.042, 0.127]
Same three inputs, refit on our folds	0.890	+0.002	0.046	+0.045 [0.012, 0.083]

ESC pretest band	NOVARION: share (observed CAD)	Diamond–Forrester: share (observed CAD)
low (<15%)	13.6% (13.5%)	5.1% (21.4%)
intermediate (15-85%)	27.1% (44.6%)	88.6% (73.6%)
high (>85%)	59.3% (98.1%)	6.2% (94.1%)

The extra measurements add discrimination beyond age, sex and chest-pain type, and NOVARION is better calibrated on this high-prevalence cohort; the published score leaves 88.6% of patients in the intermediate band, NOVARION 27.1%. The bands are descriptive only: NOVARION recommends no tests or treatment, and its low band still holds 13.5% observed CAD. **Subgroups** (sex, age; full table in `results/tables.md`): discrimination is similar by sex but lower in patients aged 65 and over (AUC CAD 0.88, LAD 0.70, LCX 0.59, RCA 0.68), with wide intervals. A TRIPOD+AI checklist maps every reporting item (`docs/tripod_ai.md`).

Robustness. A label-permutation test (1000 shuffles; 5-fold CV AUC of each deployed configuration) puts every target above all shuffled-label runs (p = 0.001 for each, the smallest attainable; null AUC ≈ 0.50 ± 0.05; "evidence against the null under the test assumptions"). When the four model families disagree on a held-out patient, the selected model errs far more often (CAD 35.1% vs 8.9%, LAD 45.6% vs 19.1%), so family disagreement is a useful descriptive cue for hard cases.

Table 4 — feature-group ablation (descriptive): ΔROC-AUC when a group is removed and the training-side procedure is re-run on the same 25 folds, logistic-regression family.

Target	Full	− Demogr.	− History	− Symptoms/exam	− ECG	− Labs	− Echo
CAD	0.932	-0.019	-0.007	-0.060	-0.004	-0.005	-0.012
LAD	0.839	-0.014	+0.007	-0.039	-0.002	-0.002	-0.037
LCX	0.714	-0.056	+0.017	-0.032	+0.007	-0.018	+0.009
RCA	0.743	-0.030	-0.020	-0.023	+0.003	-0.030	+0.005

Symptoms and examination (typical chest pain) carry the most signal for CAD and LAD; echo (RWMA region, ejection fraction) matters for LAD; demographics (age) for LCX. **Error analysis** (pooled OOF): CAD 42/303 errors, of which 8 were shown with HIGH model confidence (10 false positives fall in the HIGH-positive region; the atypical-profile rule downgrades the rest to LOW). Rule-selected cases (lowest id meeting each criterion) include ZAS-0094, the CAD high-positive-region error — which is exactly the dataset's one label-discordant record (LAD stenotic, Cath normal) — and ZAS-0004, whose atypical profile forces LOW confidence.

Target	Errors	FP	FN	Shown with HIGH confidence
CAD	42/303	12	30	8
LAD	76/303	35	41	0
LCX	101/303	73	28	0
RCA	95/303	55	40	0

5. 3D visualization, interpretability and the system

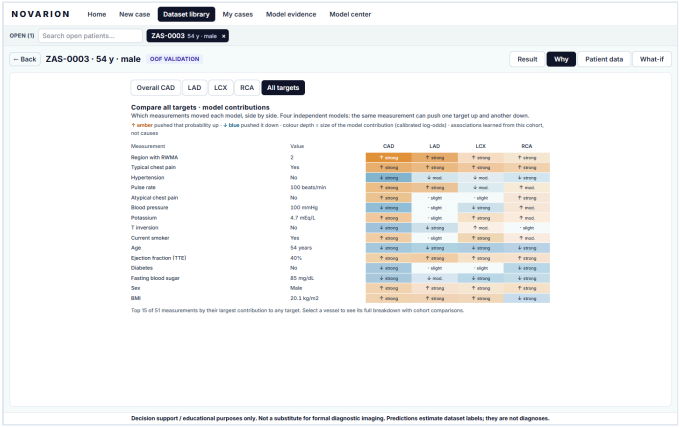


Figure 2 — Compare all targets (a case's Why tab): a features × {CAD, LAD, LCX, RCA} contribution heatmap (amber ↑ / blue ↓, depth = size); the Result tab adds matching callouts on the heart.

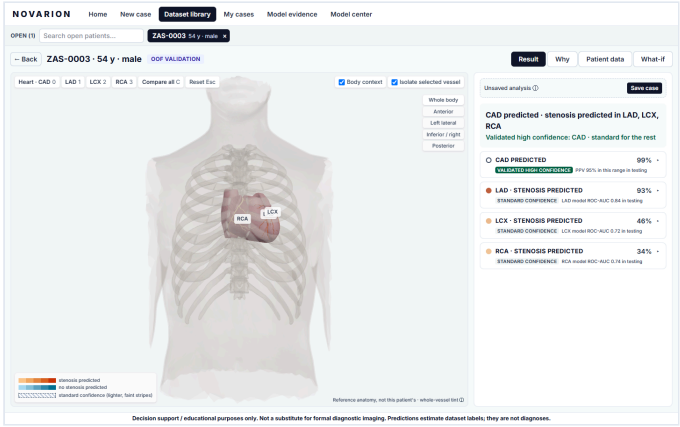


Figure 3 — Whole-body view: the heart inside a translucent torso and rib cage (body context fades automatically in vessel close-ups).

Anatomy pipeline. BodyParts3D 4.0 (CC BY 4.0): FMA concepts are resolved through the IS-A tree to element meshes and merged into named nodes (vessel_lad, vessel_lcx, vessel_rca, left main, myocardium) plus non-pickable body context (skin cropped to the trunk, rib cage); decimated to 131,990 triangles in total (2.5 MB GLB, anatomy v1.0.0) and recorded in a hash-chained anatomy manifest. A single anatomy map binds each target id to its node, camera preset and offline-computed label anchor; tests assert the correspondence. **Interaction:** rotate/zoom/pan, five camera presets, click a vessel in 3D or its card (one store action, so 3D and panels can never disagree), isolation, keyboard shortcuts, deep links, hover tooltips, compare mode. Colour is a monotone function of the calibrated probability within each predicted class (OKLCH, threshold as zero point); the whole named vessel is tinted — no lesion is implied. Rendering is on-demand, so the 3D stays responsive without a dedicated GPU. **What-if:** for a new or edited case, one input at a time is varied across its training range (age and sex locked) and the probability curve is drawn against the threshold, labelled "model sensitivity, not treatment simulation"; Unknown inputs that could flip the predicted class are listed. **Interpretability:** per-vessel contributions with strength words, presentation-group sums, a measurement table with each value's percentile or prevalence among that target's positive and negative training patients (n shown; "insufficient evidence" below 20), similar public-dataset patients, an empirical evidence ledger, and a Model Evidence page with ROC/PR/reliability/ threshold-sweep figures. **Workflow:** manual or CSV/JSON import (exact column matching, labels stripped, identifiers dropped or rejected, units declared or confirmed, server-side quality gate); live re-prediction 300 ms after an edit; cases, scenarios and immutable reviews in IndexedDB; Review Snapshot PDF/JSON built only from the stored analysis with an RFC 8785 + SHA-256 hash that also covers the embedded 3D render.

6. Governed learning loop (designed for continual supervised updating)

Predictions always come from the current production model; nothing retrains automatically and the model never learns from its own predictions. A reviewer can record an angiographic outcome on a saved case — stored as *pending*, bound to that exact input hash. An allowlisted learning package (inputs, labels, source enums; no notes, names, dates or identifiers) goes to an offline steward step (`make verify-outcomes`: label definition, provenance, inputs predating the angiogram, conflicts, duplicates). `make candidate` freezes the newest verified cases as a forward holdout before any challenger-side step, trains a challenger with the same final-model procedure, freezes its configuration and scores champion and challenger once on that holdout. A pre-registered paired gate (per target: ΔAUC 95% CI lower bound > −0.05, Brier and calibration tolerances, at least one target improved; too few holdout positives/negatives = "insufficient evidence", not failure) produces a report that a person must approve (`make promote`; `make rollback` restores any earlier immutable bundle). Every step is appended to a hash-chained audit log, and the Model Center shows registry, gate reports and log; it has no promote button. An end-to-end verify → candidate → reject → promote → rollback drill is part of the test suite.

Table 5 — historical replay (research demonstration, NOT temporal validation): the same 303 patients in 10 pre-registered random arrival orders; AUC of the model trained on n patients, scored on the next 50 arrivals (mean ± SD across orders; lr + xgboost search space).

Target	n = 153	n = 203	n = 253
CAD	0.910 ± 0.050	0.924 ± 0.044	0.922 ± 0.035
LAD	0.821 ± 0.052	0.846 ± 0.027	0.826 ± 0.037
LCX	0.671 ± 0.084	0.695 ± 0.052	0.714 ± 0.057
RCA	0.731 ± 0.050	0.752 ± 0.069	0.730 ± 0.041

Only LCX improves consistently as cases accumulate; the other targets plateau within the across-order spread. The gate promoted a challenger in 1 of 20 replayed decisions — with 50-patient holdouts the paired ΔAUC interval is wide, so the gate is conservative by design and "no promotion: the current model remains" is the usual, displayed outcome.

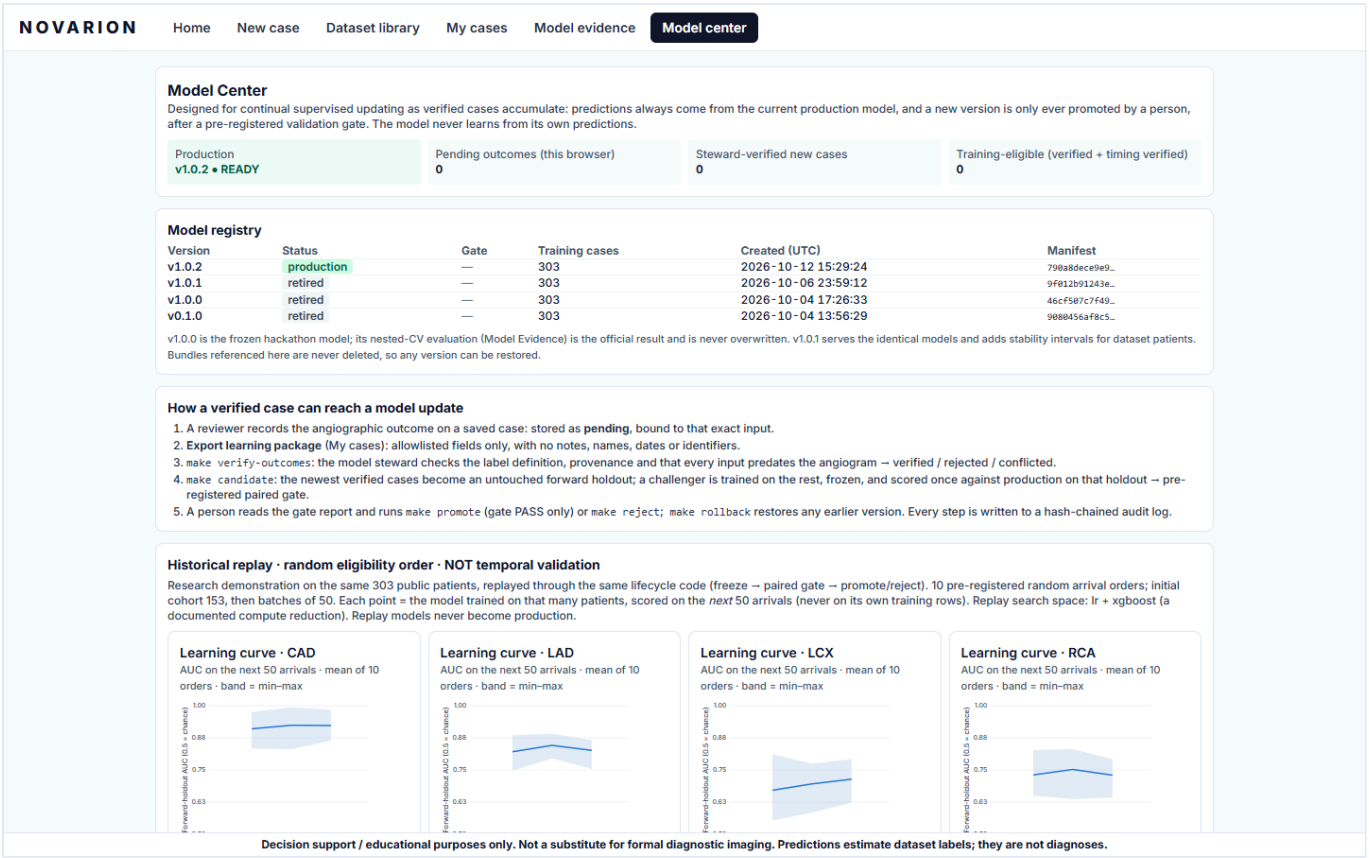


Figure 4 — Model Center (read-only): production status, registry, the governed update steps and the replay learning curves (forward-holdout AUC; band = range across arrival orders).

7. Usage, reproducibility, limitations

Run: `make reproduce` (download → train → evaluate → bundle) and `make serve` + `npm run dev`, or `docker compose up`. Seeds, the split manifest, the training-matrix hash (`3864a1ab3cd7…`) and per-file artifact hashes are recorded; the backend refuses a modified bundle. A golden prediction set guards future patch releases; demo patients are picked by an outcome-blind hash rule. Tests: 190 backend/ML (leakage, contract, additivity, calibration, thresholds, hashing, bundle integrity, golden set, stability, learning loop), 24 frontend unit tests and 50 browser end-to-end flows, passing against the real model in Edge, Chrome and Firefox (axe accessibility checks included). Measured: `/predict` p95 104 ms; 3D median 59.9 fps on Intel Iris Xe integrated graphics; backend memory 178 MB after start, 184 MB peak. If the backend is unreachable, a hash-verified static bundle keeps every dataset-patient flow working offline. **Reproduced:** a full run on a fresh clone (fresh data download) matched every fold, metric, golden prediction, stability interval and replay result exactly (passed; `results/reproduction.json`). **Limitations:** one public, single-source cohort of 303 patients with high prevalence (PPV/NPV do not transfer); internal validation only; weaker performance in patients aged 65 and over; contributions are associations, not causes; the anatomy is a reference model. **Credits:** dataset UCI #411 (CC BY 4.0); anatomy BodyParts3D, © The Database Center for Life Science (CC BY 4.0).